Sana Raza

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sana-raza-engineer | Sanarazaaa

Education

- **Louisiana State University [🌐]** *Jan 2026 - Present*
  PhD in Biological Sciences Baton Rouge, US
- **NED University of Engineering and Technology [🌐]** *Nov 2017 - Oct 2021*
  Bachelor of Biomedical Engineering Karachi, Pakistan
  - CGPA: 3.47/4.00

Projects

- **Species Identification and Structural Characterization**
  Tools: SWISS-MODEL, BLAST, Click2Drug, PubMed [🌐]
  - Analyzed two unknown protein sequences (SEQ1 and SEQ2) for species identification and drug discovery applications using bioinformatics tools
  - Used SWISS-MODEL for structure modeling, BLAST for sequence homology, Click2Drug for drug design, and PubMed for literature review
  - Identified a kinase family protein, suggesting potential for cancer therapy development

- **Neural Oscillation and Burst Firing Simulation**
  Methods: Conductance-based modeling, FFT, Hilbert transform [🐙]
  - Developed a computational model to simulate neuron firing dynamics under oscillatory inputs (theta and gamma rhythms)
  - Implemented membrane potential updates, spike detection, and burst analysis using conductance-based equations
  - Applied the Fast Fourier Transform (FFT) for resonance analysis and visualized dominant neural frequencies
  - Conducted phase-locking analysis using Hilbert transforms to study spike synchronization with theta oscillations

- **Cogniquest**
  Interactive cognitive test [🌐]
  - Developed an interactive cognitive test to assess working memory (Digit Span Task) and problem-solving skills (math exercises)
  - Engaged users in tasks of increasing difficulty while collecting performance data to evaluate accuracy and progression

- **Studying Psychological Impact via Structural Equation Modeling (Final Year Project)**
  Tools: SPSS, Structural Equation Modeling (SEM) [🌐]
  - Determined the prevalence of psychological well-being among the Pakistani public and identified primary predictors of distress
  - Studied the relationships between mental health factors and their impact on mental well-being using SPSS
  - Analyzed structural relationships between measured and latent variables using SEM

Honors and Awards

- **Fully Funded Research Scholarship**
  Selected by a UC Irvine professor [🌐]
  ◦ Selected for a highly competitive, fully funded research scholarship
- **Academic Rank: 5th of 69**
  NED University of Engineering and Technology
  ◦ Ranked 5th in a class of 69 students
- **1st Position, Harvard CS50 Puzzle Day** *2024*
  Harvard University [🌐]
  ◦ Solved 9 out of 9 puzzles
- **Advent of Code** *2024*
  Advent of Code [🌐]
  ◦ Came second on the local leaderboard and ranked in the top 100 of the global leaderboard
- **Competition Achievement: CALICO** *Nov 2024*
  CALICO, University of California, Berkeley [🌐]
  ◦ Ranked among the top 15 teams on the national leaderboard
  ◦ Solved 8 out of 11 programming challenges within a 3-hour time frame

Leadership Experience

- **Team Lead, DesignHand** *May 2024*
  Hello GPT-4o AI Challenge, lablab.ai [🌐]
  ◦ Led the team in developing a web-based app that transformed raw rooms into aesthetically pleasing spaces based on user-provided images and text
  ◦ Designed an intuitive interface that allows users to easily visualize and implement suggested design changes
- **Frontend Lead, Content Generator and Summarizer** *Dec 2024*
  BuildwithAI Hackathon, GENAI.WORKS
  ◦ Spearheaded the frontend development and crafted an intuitive user interface focused on a seamless user experience
  ◦ Guided the team in delivering a polished and cohesive frontend, aligning design and functionality with the overall project goals

Volunteer Experience

- **Python Trainer** *Oct 2024 - Dec 2024*
  iCode Guru, Remote [🌐]
  ◦ Adapted training materials to fit the needs of participants, including housewives, students, and professionals
  ◦ Taught students the basics of Python and programming while ensuring class participation
- **Maths Trainer, MIT Emerging Talent Program Entry Exam** *Oct 2024*
  iCode Guru, Remote [🌐]
  ◦ Taught foundational math topics like vectors, matrices, and scalars in easily digestible language
  ◦ Used problem-solving exercises to solidify learners' understanding of mathematical principles in technical contexts
- **IELTS Trainer** *May 2024 - Aug 2024*
  iCode Guru, Remote [🌐]
  ◦ Taught underprivileged students tips and methods for the IELTS
  ◦ Individually trained students to overcome mistakes in the reading and speaking modules, and gave feedback on writing tasks
- **Fundraising Volunteer** *Oct 2019 - Sep 2021*
  FalaheBashar, Karachi, Pakistan [🌐]
  ◦ Helped with data entry, record-keeping, and other administrative tasks related to fundraising activities
  ◦ Assisted with online fundraising efforts, including crowdfunding and virtual events